

ACHINTYA RAI

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EDUCATION

University of California, San Diego
Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028
GPA: 3.9/4.0

WORK EXPERIENCE

Adobe
Software Engineering Intern

San Jose, CA
June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks, isolating **86,387** duplicates and a **47%** null rate.
- Engineered SQL and MLOps pipelines for weekly retraining and hourly inference on ML models.
- Designed a time-series anomaly detector, benchmarking statistical algorithms to select an ML model handling multi-layer seasonality across **10+** workflows and **30+** countries.

VALT Health
Founding Software Engineer

Remote
Aug. 2025 – Feb. 2026

- Engineered a Flask backend for **100k+** datapoints, unifying formats across **20+** device sources.
- Implemented asynchronous batch processing and storage workflows in GCP, yielding a **35%** increase in execution speed and **60%** reduction in redundant database write operations.
- Automated environment setup and data seeding via Bash scripts, cutting testing time up to **80%**.

Decklar (formerly Roambee)
Software Engineering Intern

Sunnyvale, CA
June 2024 – Aug. 2024

- Developed real-time shipment monitoring for **Amazon, Coca-Cola, and Samsung**, optimizing Python and NumPy data pipelines for a **15%** capacity boost across **1M+** weekly events.
- Implemented advanced data structures for analytics modules, yielding **20%** faster reporting.
- Enhanced workflow efficiency **15%** via Panels-based UI refinements and iterative user testing.

RESEARCH

Graph Theory & Neural Architecture
Undergraduate Research Fellow (NSF REU)

San Diego, CA
March 2026 – Present

- Engineered a Python pipeline to analyze LLM embedding manifolds (Llama-2, DeepSeek-V2), implementing weight-matrix extraction to compress **500GB** checkpoints to **13GB** localized manifolds.
- Optimized placement scripts and normalized preprocessing for a **30x runtime reduction** (30 min to <1 min), enabling large-scale manifold analysis on resource-constrained **8GB-RAM** systems.
- Applied **Rent's Rule** and Louvain clustering to raise ICN density from **0.18%** to **13.25%**.

PROJECTS

NanoGPT | *Python, PyTorch, Transformers, Neural Networks* | 

Mar. 2026

- Implemented a transformer from scratch in **Python/PyTorch**, building multi-headed attention and backpropagation for a character-level model with **100k+** parameters.
- Engineered scalable dataloaders to stream tokenized datasets at **100%** hardware utilization.

SquaredUpChess | *React, TypeScript, Django, Websockets, AWS, Coolify, Docker* | 

Aug. 2025 – Dec. 2025

- Architected **20+ Django REST endpoints** with tuned SQLite indexing for low-latency updates.
- Established a **CI/CD pipeline** with error logging and automated testing for **99.9%** reliability.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, TypeScript, JavaScript, SQL (Postgres/SQLite), Bash, HTML/CSS

Development: React, Django, Flask, FastAPI, Node.js, Asyncio, WebSockets, JUnit, PyTest

AI & Cloud: PyTorch, TensorFlow, AWS, GCP, Docker, Linux, GitHub Actions, MCP, Gemini/OpenAI API